False and multicycle paths (engine view)

Exceptions change which arcs propagate and how large the required window is

Goldens to remember

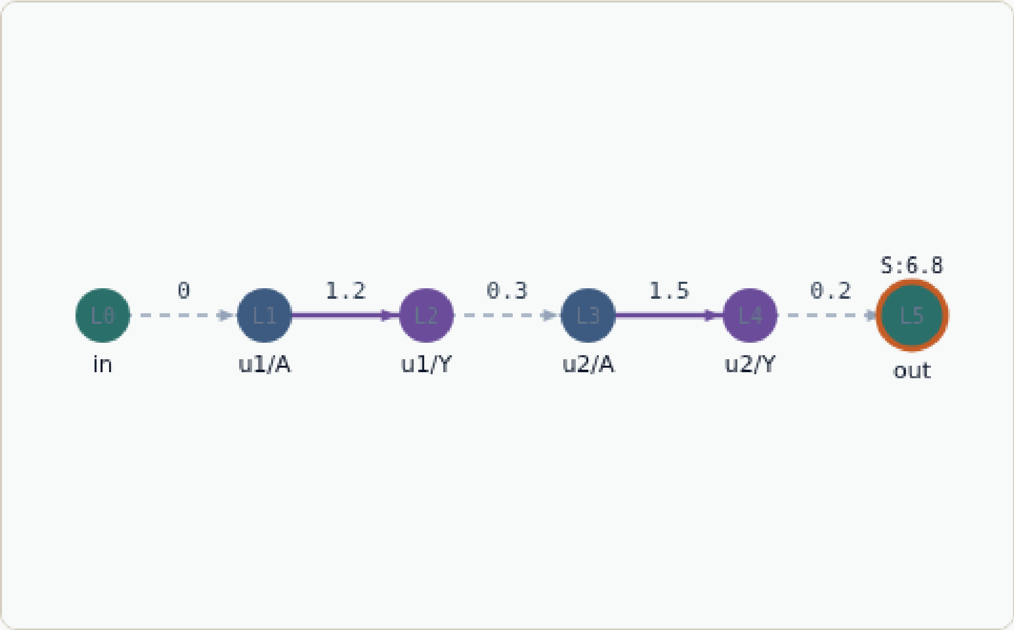
- Multicycle cycles=2 → R(out)=20, slack=16.8
- False-path disables u1/Y→u2/A so A(u2/A) falls to 0 in the lite model
- Keep these numbers handy, the browser challenges and Track A tests use the same instance
- <!-- algorithm-walkthrough -->

Normal single-cycle setup

FALSE AND MULTICYCLE PATHS

Normal single-cycle setup

Step 1 / 5 · normal-check · module03-03-false-multicycle-lite



Explanation

Without exceptions, required at out is one period (10) and setup slack is 6.8. That is the baseline the exceptions will change.

- cycles = 1
- R(out)=10
- slack=6.8

```
setup slack=6.8
```

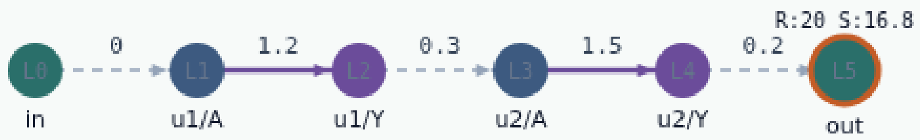
Normal single-cycle setup

Multicycle widens the required window

FALSE AND MULTICYCLE PATHS

Multicycle widens the required window

Step 2 / 5 · multicycle · module03-03-false-multicycle-lite



Explanation

A setup multicycle of 2 means required = 2 × period = 20. Slack becomes 16.8. The graph did not change—only the endpoint budget did.

- R = period × cycles
- cycles=2 → R=20
- slack=16.8

```
required out=20
setup slack=16.8
```

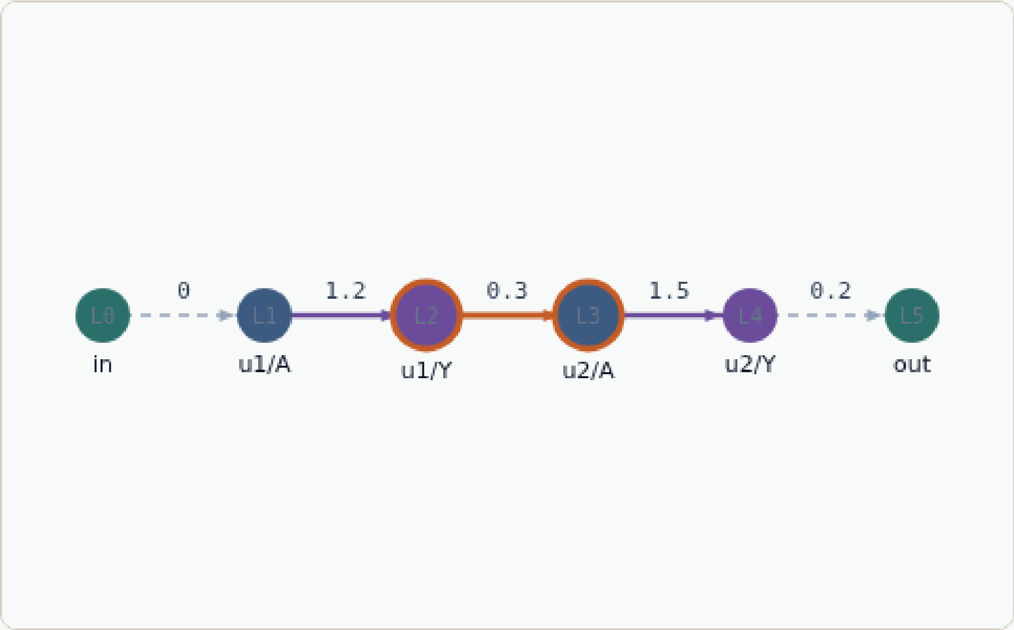
Multicycle widens the required window

False path disables an arc

FALSE AND MULTICYCLE PATHS

False path disables an arc

Step 3 / 5 · false-path · module03-03-false-multicycle-lite



Explanation

Marking the bridge net u1/Y → u2/A as false removes it from propagation. Downstream pins no longer see the real wavefront—u2/A falls back to 0 in this lite engine.

- Disable arc u1/Y|u2/A
- Propagation skips it
- Lite model: orphan pin → 0

```
disabled: u1/Y|u2/A
A(u1/Y)=1.2
A(u2/A)=0
```

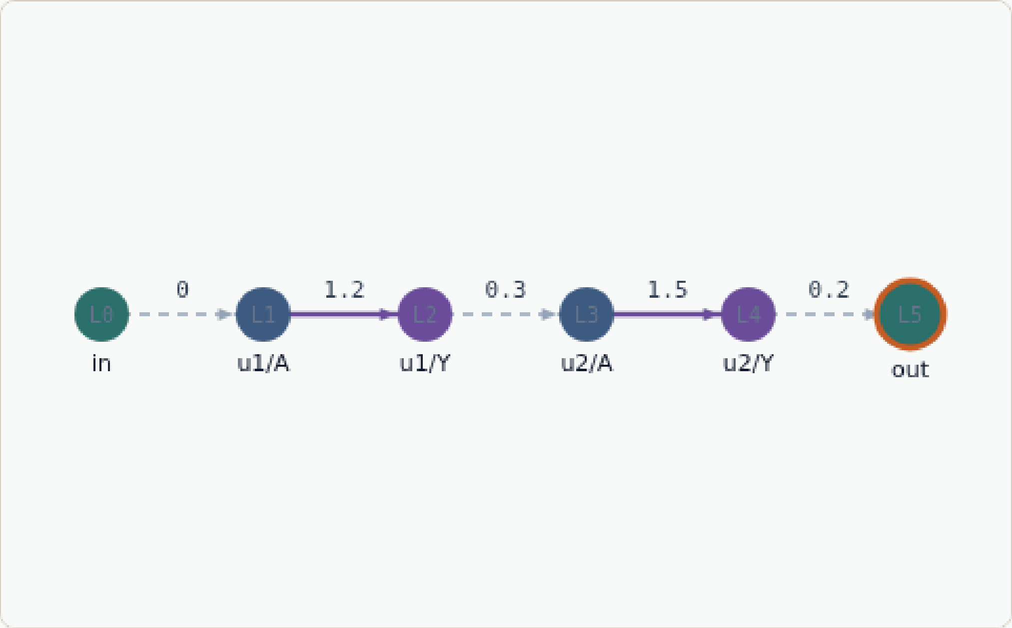
False path disables an arc

Exceptions are engine data

FALSE AND MULTICYCLE PATHS

Exceptions are engine data

Step 4 / 5 · engine-data · module03-03-false-multicycle-lite



Explanation

SDC authoring lives in learn_sdc. Here you only consume false-path and multicycle as flags the timer reads—same idea as production engines.

- Constraints → engine inputs
- Not GUI click-paths
- Wrong exception → wrong slack

```
false-path: remove arc
multicycle: scale required
```

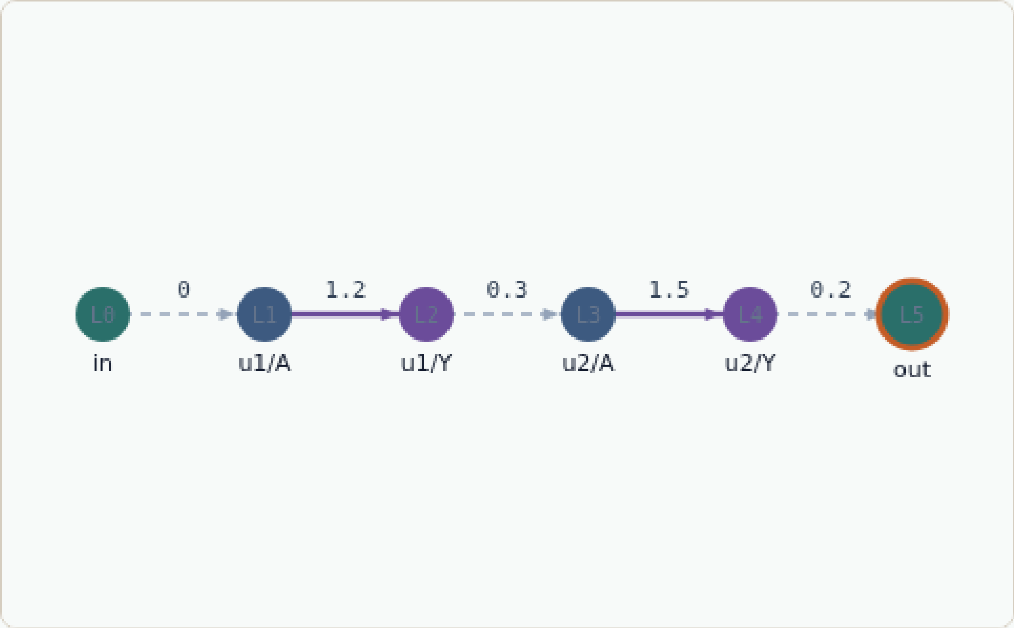
Exceptions are engine data

Always recheck after exceptions

FALSE AND MULTICYCLE PATHS

Always recheck after exceptions

Step 5 / 5 · recheck · module03-03-false-multicycle-lite



Explanation

Apply exceptions, then recompute tags and slack. Never keep a stale 6.8 after a multicycle or false-path change.

- Apply → recompute → report
- Stale tags lie
- Next: offline compare habit

```
normal 6.8
multicycle 16.8
false-path breaks the bridge
```

Always recheck after exceptions

Browser lab track

- In the browser lab, open **false-multicycle-lite**
- Load the starter, run the analysis once, and read the metrics panel
- Orient yourself, challenge panel, canvas, Check, then mirror the same goldens in code

Implement track

- In the implement track
- Run `python3 common/test_propagate.py` (and the timing-graph test) until the goldens print

Pitfall

- Do not mix setup and hold required maps
- Do not propagate before the graph is levelized
- After an edit or exception, recompute, stale tags lie

Your turn

- Finish the checklist on at least one track, preferably both
- When your numbers match the goldens, take the quiz, then continue